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DETAIL SPECIFICATION

CARTRIDGE, 40 MILLIMETER, PROJECTILE ASSEMBLY, PRACTICE - M385A1

Inactive for new design after 12 February, 1996

This specification is approved for use by U.S. Army Armament Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Cartridge, 40 Millimeter - Projectile Assembly, Practice - M385A1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806 5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 1310

Distribution Statement D. Distribution authorized to the Department of Defense and U.S. DoD contractors only to protect, Critical Technology, as determined 1 April 2008. Other requests shall be referred to the Office of the Project Manager for Maneuver Ammunition Systems, ATTN: SFAE-AMO-MAS-SMC, Picatinny, NJ 07806-5000

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

8866714	Projectile Assembly, For Cartridge, 40MM: Practice, M385A1
8866715	Body
8886326	Cartridge, 40MM: Practice, M385A1

(Copies of these drawings may be requested online at pica.drawing.request@conus.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B117	Standard Practice for Operating Salt Spray (Fog) Apparatus

(Copies of this document are available online at http://www.astm.org or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959)

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2.4 Order of precedence. Unless otherwise specified in this document or in the contract, in the event of a conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.1.2 Conformance. When specified, a sample shall be subjected to conformance inspection in accordance with 4.4

3.2 M385A1 projectile assembly and components.

3.2.1 Body. The body shall be in accordance with drawing 8866715.

3.2.2 Projectile assembly. The projectile assembly shall be in accordance with drawing 8866714.

3.2.2.1 Push test. The push test of rotating band shall be in accordance with drawing 8866714.

3.2.2.1.1 Change in knurling configuration. Every time there is a change in the knurling configuration, a push test of rotating band on projectile assembly First Article Test shall be preformed.

3.3 Coating thickness. The thickness of the anodic coating shall be a minimum of 0.3 mil (.0003 inches).

3.4 Markings. Marking and stamping shall be neat, sharply define, and durable in accordance with drawing 8866714.

3.5 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.6 Workmanship. All parts and assemblies shall be free of burrs, cracks, voids, pipes, inclusions, laminations chips, sharp edges, unblended radii, surface defects, dirt, grease, rust, corrosion products, other foreign matter and, any other material defect. The cleaning method used shall not cause defect to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coating shall be continuous; however, a few light scratches not exposing base material may be permitted.

4. VERIFICATION

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TABLE I. Requirement/verification cross reference matrix

Method of verification N/A - Not Applicable 1 - Analysis 2 - Demonstration 3 - Examination 4 - Test							Classes of verification A - First Article B - Conformance		
SECTION 3 REQUIREMENT	DESCRIPTION	VERIFICATION METHODS					VERIFICATION CLASS		SECTION 4 VERIFICATION METHOD
		N/A	1	2	3	4	A	B	
3.1.1	FIRST ARTICLE				X	X	X		4.3
3.1.2	CONFORMANCE				X	X		X	4.4
3.2	PROJECTILE ASSEMBLY & COMPONENTS				X	X	X	X	4.3, 4.4
3.2.1	BODY				X	X	X	X	4.4.2.1, 4.5.1
3.2.2	PROJECTILE ASSEMBLY				X	X	X	X	4.4.2.2
3.2.2.1	PUSH TEST				X	X	X	X	4.5.2
3.2.2.1.1	CHANGE IN KNURLING CONFIGURATION				X	X	X	X	4.5.2.1, Table IV
3.3	COATING THICKNESS				X	X	X	X	4.5.3
3.4	MARKINGS				X		X	X	4.5.4
3.5	AMMUNITION LOT NUMBERING				X		X	X	4.5.5
3.6	WORKMANSHIP				X		X	X	4.4.2.1, 4.4.2.2

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection Condition. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First Article. When specified, a sample of the Projectile Assembly, 40MM: Practice, M385A1 shall be subjected to first article inspection in accordance with Table II and Table III.

4.3.1 First Article Quantity. The first article sample shall consist of the following items in sample quantities as indicated in Table II and Table III.

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TABLE II. First article quantity

PART DESCRIPTION	DRAWINGS	QUANTITY
Body	8866715	60
Projectile Assembly	8866714	100

4.3.2 Inspections to be performed. The first article inspection shall be 100% examination of all drawing requirements including workmanship for the parts and assemblies listed and the tests specified in Table III herein.

4.3.3 First Article rejection. If any item of the sample fails to comply with the First Article requirements, the sample shall be rejected.

TABLE III. First article inspection

EXAMINATION OR TEST	NO. SAMPLE UNITS	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Body (Dwg. 8866715)			
Examination for defects	50 1/	3.2.1	Gage
Salt spray	10	3.2.1.1	4.5.1
Projectile Assembly (Dwg. 8866714)			
Examination for defects	50 1/	3.2.2	Gage
Push test of rotating band	50	3.2.2	4.5.2, 4.5.2.1, Table IV
Notes: 1/ 25 samples shall be inspected for unlisted characteristics.			

4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. Each inspection lot of projectiles shall be produced with components complying with the following constraints:

a. Metal parts or components from one lot interfix number, from one manufacturer.

4.4.1.1 Ammunition lot numbers. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168. Visually verify that each lot is identified as the type, caliber, model, and lot number in accordance with MIL-STD-1168.

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4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major, and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors, and Inspection Level II for all characteristics defined as Minors. Inspection Level VII shall be used for all characteristics defied as Criticals in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 and 4.4.2.2. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE	SHEET 1 OF 2		DRAWING NUMBER 8866715
4.4.2.1	Body			NEXT HIGHER ASSEMBLY 8866714
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101	Diameter of rotating band surface (after knurl)	Level IV	3.2.1	Gage
102	Spherical radius	Level IV	3.2.1	Gage
103	Location of basic diameter (excluding cut-off nib)	Level IV	3.2.1	Gage
104	Run-out of indicated area on taper with body diameter	Level IV	3.2.1	Gage
105	Body diameter	Level IV	3.2.1	Gage
106	Diameter between knurl	Level IV	3.2.1	Gage
107	Width of knurl crest	Level IV	3.2.1	Gage
108	Width of rotating band surface	Level IV	3.2.1	Gage
109	Length from back of rotating band surface to front end of nose	Level IV	3.2.1	Gage
110	Location of bourrelet shoulder	Level IV	3.2.1	Gage
111	Diameter of bourrelet	Level IV	3.2.1	Gage
112	Diameter below bourrelet	Level IV	3.2.1	Gage
113	Salt spray 1/	2 per 8 hr shift	3.2.1	4.5.1
114	Coating thickness	Level IV	3.3	4.5.3
MINOR				
201	Protective coating with bare spot on bottom or forward of rotating band groove	Level II	3.2.1	Visual
202	Incorrect color	Level II	3.2.1	Visual
203	Evidence of poor workmanship	Level II	3.6	Visual

Notes:
1/ The production represented by the 8-hour shift sample shall be rejected upon non-conformance to the acceptance criteria defined in MIL-A-8625 when salt spray tested in accordance with 4.5.1.

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Conformance inspection by classification of characteristics

PARAGRAPH	TITLE	SHEET 1 OF 1		DRAWING NUMBER 8866714
4.4.2.2	Projectile Assembly			NEXT HIGHER ASSEMBLY 8886326
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101	Profile of rotating band	Level IV	3.2.2	Gage
102	Diameter of rotating bands	Level IV	3.2.2	Gage
103	Diameter of large groove	Level IV	3.2.2	Gage
104	Diameter of small groove	Level IV	3.2.2	Gage
105	Distance from nose to back of rotating band	Level IV	3.2.2	Gage
106	Total run-out of rotating bands with bourrelet and body diameter	Level IV	3.2.2	Gage
107	Total length (excluding cut-off nib)	Level IV	3.2.2	Gage
108	Push test of rotating band	80 1/	3.2.2	4.5.2, 4.5.2.1, Table IV
MINOR				
201	Marking missing	Level II	3.2.2	Visual
202	Evidence of poor workmanship	Level II	3.6	Visual

Notes:
1/ Accept on zero (0) and reject on one (1) or more failures.

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4.5 Methods of inspection.

4.5.1 Salt spray. The specified part shall be subjected to the salt spray in accordance with ASTM B117 / MIL-A-8625. Testing shall be conducted for 336 hours using a salt solution of 5 percent. The sample shall be rejected if the test results fail to meet the criteria as specified in MIL-A-8625. This is a destructive test, parts shall not be returned to the lot.

4.5.2 Push test of rotating band on projectile assembly. The projectile assembly shall be placed in a fixture and the force specified on Dwg. 8866714 shall be applied. Observation shall be made for the requirement specified. This test is a destructive test. Parts tested shall not be returned to the lot. Deformation of the chamfered leading edge of the rotating band caused by compression in the test fixture is permissible. Movement as specified on Dwg. 8866714 is defined as movement of the back face of the rotating band relative to the body.

4.5.2.1 Change in knurling configuration. The body assemblies shall be subjected to temperature conditioning testing. For the ambient portion of the test, the bodies shall be conditioned at 70 ± 5 F at least eight (8) hours, and then tested within fifteen (15) minutes after being removed from the temperature soak. For the high temperature test, the body assemblies shall be conditioned at 125 ± 5 F at least four (4) hours, and then tested within fifteen (15) minutes after being removed from the temperature soak. For the low temperature test, the body assemblies shall be conditioned at - 25 ± 5 F at least four (4) hours, and then tested within fifteen (15) minutes after being removed from the temperature soak. Fifty (50) body assemblies shall be tested for each hot, cold and ambient portions of this test. Failure to meet the requirement on the specified drawing shall be the reason for rejection of the First Article. See Table IV.

TABLE IV. Ballistic testing consideration of an alternative knurling configuration

ROUNDS	TEMPERATURE	TEST	RANGE	INSPECTION
144	Ambient	Single shot 1/ 4/	200 feet	Recover and inspect
50	Hot	Single shot 1/ 4/	200 feet	Recover and inspect
50	Cold	Single shot 1/ 4/	200 feet	Recover and inspect
96	Ambient	Rapid fire 2/ 3/ 4/	200 feet	(5-6 round bursts) recover and inspect after each burst

Notes:
1/ Single shot rounds must be recovered and inspected for a missing rotating band or movement of the rotating band and to ensure that the engraving does not cut into the aluminum knurling. Rounds should be recovered frequently to avoid the possibility of hitting each other.
2/ Rapid fire rounds must be recovered and inspected for movement of the rotating band and to ensure that the engraving does not cut into the aluminum knurling. Rounds should be recovered after each burst.
3/ Any gun stoppage during the rapid fire must have the cause determined and recorded.
4/ Accept/reject criteria on the ballistic testing is accept on 0, reject on 1 for rotating band movement, rotating band missing or engraving cutting into aluminum knurling .

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4.5.3 Anodic coating thickness. The equipment used to measure the thickness shall be according to the test methods defined in MIL-A-8625. Alternate method shall be a permascope, type EC or equivalent and a measuring probe, type Ta or equivalent. Permascope method is a non-destructive test and parts tested may be returned to the lot.

4.5.4 Markings. Visually verify the markings to meet the requirement in 3.4.

4.5.5 Ammunition lot numbering. Visually verify that an ammunition lot number has been applied to each lot as described in MIL-STD-1168.

5. PACKAGING

5.1 For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's, automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the requirements, examinations and tests for the Projectile Assembly for the Cartridge, 40MM, Practice, M385A1. There is no known commercial use, making the item military unique.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Requirements for submission of first article sample.
c. Requirement for submission of design verification sample, if applicable.
d. Requirement for submission of inspection equipment designs.
e. Requirement for submission of conformance inspection sample, if applicable.
f. Applicable national stock number.
g. Packaging requirements, if other than specified in Section 5.
h. Serialization requirements, if applicable.
i. Certificate of conformance for each lot or shipment of product, if applicable.
j. Government-Furnished Material or equipment, if applicable.
k. Provisions for critical characteristics controls.

6.3 Submission of inspection equipment designs for approval. Contractor should submit equipment designs as required to Commander, ARDEC, ATTN: AMSRD-AAR-QEM-D,

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Picatinny, NJ 07806-5000. Request letter of submittal should state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph. Inspection equipment, inspection method and the frequency of inspection should be submitted for approval to ARDEC, Attn: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000.

6.4 Contractor acceptance inspection equipment (AIE). Provision concerning the contractor's AIE used to verify the requirements of this specification should be specified in the contract.

6.4.1 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is listed in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. Inspection equipment used for acceptance of product should be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected. Any automated acceptance inspection equipment (AAIE) prove-out should be performed in accordance with MIL-A-70625.

6.5 Ammunition lot numbers. Ammunition should have ammunition data cards in accordance with MIL-STD-1168.

6.6 Submission of alternative conformance provisions. All contractor proposed alternative conformance provisions should be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.8 Subject term (key word) listing.

40MM Ammunition
M16A2 Metallic Link
M169 Cartridge Case
MK19 Mod 3 Grenade Machine Gun

Preparing activity:
Army-AR
(Project 1310-2008-014)

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